

ENGINEERING PROJECT PORTFOLIO

Smart Warehouse AMR Fleet Coordination System

Edge-AI Multi-Agent Swarm, Distributed Traffic Management & 3D Digital Twin

Candidate Name:	[Candidate Full Name]
Roll / Reg No:	[Registration / Roll Number]
Program & Dept:	[Degree / Program Name], Department of Computer Science & Robotics
Institution:	[University / Institution Full Name]
Primary Platform:	MATLAB R2021a - R2026a (OOP & 3D Digital Twin) Python 3
Date & Status:	September 2026 Verified & Final Submission Ready

TECHNICAL SKILLS MATRIX

Domain / Category	Applied Skills & Technical Competencies
Languages & Tools	MATLAB (OOP, Vectorization, Graphics Engine), Python, Git
Robotics & Navigation	A* Shortest-Path Planning, Manhattan Heuristics, Dynamic Avoid-Masks, Kinematics
Swarm Intelligence	Master-Slave Exclusive Aisle Leases, Lateral Collision Dispersal, Distributed Allocation
Networking Protocols	Peer-to-Peer (R2R) Mesh, Cross-Zone Task Trading ("Bring My Item"), 1-hop Handshake
Energy Management	Battery BMS (<30%), Task Memory Freezing (PausedForCharging), Dual-Port Balancing
Digital Twin & GUI	2D Quad-Shell Telemetry Dashboard, Interactive UITable, 3D Digital Twin with 360° Orbit

PROJECT DOSSIER: CORE ENGINEERING & ARCHITECTURE

At-A-Glance Specification

Grid & Dimensions	30×30 Grid Warehouse (900 Cells), 8 Shelves (A-H), 2 Pack Hubs, 2 Dual Chargers
Fleet Configuration	5 Autonomous Mobile Robots (AMRs: R1-R5) with differential kinematics & LiDAR puck
Core Protocols	Master-Slave Picking Leases, Lateral Corridor Dispersal, P2P Task Trading, BMS
Role in Project	Lead Robotics, Swarm Intelligence & Simulation Engineer (Full Architecture & Code)

Problem Statement & Architecture

High-density automated fulfillment centers suffer from corridor deadlocks, multi-robot picking congestion in narrow aisles, and excessive cross-facility transit overheads. Monolithic centralized servers introduce single-point failure risks and computational lag, while typical charging routines drop active customer orders. This project implements a decentralized multi-agent AMR system that eliminates deadlocks in under 0.8 seconds, autonomously trades cross-zone tasks, and preserves active tasks with zero order loss.

Core Algorithmic Innovations

- 1. A* Pathfinding with Zero-Stall Next-Step Initialization:** Evaluates Manhattan cost $f(n) = g(n) + h(n)$ with dynamic avoid-masks. Initializes *PathIndex* = 2 when starting at current position to eliminate 1-tick stationary hesitation.
- 2. Master-Slave Exclusive Aisle Leases:** Prevents congestion in narrow shelf picking aisles. The nearest robot secures the Master lease, while contender robots wait in safe buffer cells outside the aisle until the Master exits.
- 3. High-Speed Lateral Corridor Dispersal:** Detects conflicts within 3.0 cells. The higher-priority AMR (or low-battery AMR) maintains right-of-way, while the Slave executes a lateral sidestep into a parallel lane in < 0.8s.
- 4. P2P Task Trading Mesh ("Bring My Item"):** AMRs in opposing zones execute a mutual wireless handshake to swap cross-zone delivery tasks, saving 28 to 52 grid cells of travel per transaction.
- 5. Low-Battery BMS (<30%) & Task Memory Freezing:** Active orders are frozen in memory as *PausedForCharging*. Robots route to dual-port docks without corridor freezing and automatically resume interrupted jobs post-charge.

RESULTS, TELEMETRY & REPOSITORY NAVIGATION

Empirical Benchmark Results (500s Simulation Run)

Performance Metric	Centralized Baseline	Proposed Distributed Swarm	Empirical Gain
Order Fulfillment Throughput	18.2 orders / 500s	24.4 orders / 500s	+ 34.1% Higher
Cross-Facility Travel Distance	1,420 grid cells	1,040 grid cells	- 26.8% Saved
Corridor Conflict Resolution Time	4.2 seconds	< 0.8 seconds	81.0% Faster
Aisle Deadlocks & Freezes	14 deadlock events	0 (Zero Deadlocks)	100% Resolved
Task Loss on Low Battery	3 orders abandoned	0 (100% resumed)	Zero Task Loss

Dual Telemetry & Visualizer Environments

- **2D Telemetry Dashboard (WarehouseDashboard.m & SmartWarehouse2D_AllInOne.m):** Features a 60% floor canvas with live AMR position markers, dotted route trajectories, and four live panels: Shell 1 (System KPI HUD), Shell 2 (Scrollable R2R Dialogue Terminal), Shell 3 (Real-Time In-Hand Parcel Monitor), and Shell 4 (Interactive Orders UITable with barcodes and timestamps).
- **3D Digital Twin Visualizer (WarehouseDashboard3D.m & run_simulation_3d.m):** Renders multi-tier industrial storage racks, AMRs with rotating LiDAR pucks, **physically loaded 3D parcel boxes** on cargo decks, illuminated dual-state charging docks, glowing cyan wireless laser beams, and interactive 360° orbital camera controls.

Repository Execution Quick-Start

Command	Script File	Functional Responsibility & Mode
run_simulation(1)	run_simulation.m	Launches nominal 2D modular fleet simulation with full telemetry.
run_simulation_3d	run_simulation_3d.m	Launches 3D Digital Twin simulation with 360° camera orbit & parcel boxes.
SmartWarehouse2D_AllInOne	SmartWarehouse2D_AllInOne.m	Runs standalone single-file 2D simulation with zero folder dependencies.
run_simulation(5)	benchmarks/	Executes full 13-point automated unit & integration test suite (100% Pass).